

Research Notes on Foundation model AI

Foundation model AI

>> Section 1

Release strategies After a foundation model is built, it can be released in one of many ways. There are many facets to a release: the asset itself, who has access, how access changes over time, and the conditions on use. All these factors contribute to how a foundation model will affect downstream applications. In particular, the two most common forms of foundation model release are through APIs and direct model downloads. When a model is released via an API, users can query the model and receive responses, but cannot directly access the model itself. Comparatively, the model could be directly downloadable for users to access and modify. Both release strategies are often classified as an open release. The exact definition of an open release is disputed, but widely accepted requirements are provided by the Open Source Initiative. Some open foundation models are: PaLM 2, Llama 2, Granite, and Mistral. While open foundation models can further research and development more easily, they are also more susceptible to misuse. Open foundation models can be downloaded by anyone, and particularly powerful models can be fine-tuned to intentionally or unintentionally cause harm. During a closed release, the foundation model cannot be accessed by the public, but is used internally by an organization. Such releases are considered safer, but offer no additional value to the research community or the public at large. Some foundation models like Google DeepMind's Flamingo are fully closed, meaning they are available only to the model developer; others, such as OpenAI's GPT-4, are limited access, available to the public but only as a black box; and still others, such as Meta's Llama 2 are open, with broadly available model weights enabling downstream modification and scrutiny.

>> Section 2

In the United States, the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence defines a foundation model as "an AI model that is trained on broad data; generally uses self-supervision; contains at least tens of billions of parameters; is applicable across a wide range of contexts". In the United States, the proposed AI Foundation Model Transparency Act of 2023 by House Representatives Don Beyer (D-VA) and Anna Eshoo (D, CA) defines a foundation model as "an artificial intelligence model trained on broad data, generally uses self supervision, generally contains at least 1,000,000,000 parameters, is applicable across a wide range of contexts, and exhibits, or could be easily modified to exhibit, high levels of performance at tasks that could pose a serious risk to security, national economic security, national public health or safety, or any

combination of those matters." In the European Union, the European Parliament's negotiated position on the E.U. AI Act defines a foundation model as an "AI model that is trained on broad data at scale, is designed for generality of output, and can be adapted to a wide range of distinctive tasks". In the United Kingdom, the Competition and Markets Authority's AI Foundation Models: Initial Report defines foundations model as "a type of AI technology that are trained on vast amounts of data that can be adapted to a wide range of tasks and operations." The United States's definitions are the only ones to make reference to the size of a foundation model, and differ on magnitude. Beyer and Eshoo's definition also specifies that foundation models must achieve a level of performance as to be a potential danger. In contrast, the E.U. definition requires the model to be designed for generality of output. All definitions agree that foundation models must be trained on a broad range of data with potential applications in many domains.

>> Section 3

Modeling For a foundation model to effectively generalize, it must acquire rich representations of the training data. As a result, expressive model architectures that efficiently process large-scale data are often preferred in building foundation models. Currently, the Transformer architecture is the de facto choice for building foundation models across a range of modalities.

>> Section 4

Data Foundation models are trained on a large quantity of data, working under the maxim "the more data, the better." Performance evaluation does show that more data generally leads to better performance, but other issues arise as data quantity grows. Tasks like managing the dataset, integrating data across new applications, ensuring adherence to data licenses, and maintaining data quality all become more difficult as data size grows. The specific demands of foundation models have only exacerbated such issues, as it remains the norm for large foundation models to use public web-scraped data. Foundation models include also search engines data and SEO meta tags data. Public web data remains a plentiful resource, but it also demands stringent moderation and data processing from foundation model developers before it can be successfully integrated into the training pipeline. Training foundation models often runs the risk of violating user privacy, as private data can be disclosed, collected, or used in ways beyond the stated scope. Even if no private data is leaked, models can still inadvertently compromise security through learned behavior in the resulting foundation model. Data quality is another key point, as web-scraped data frequently contains biased, duplicate, and toxic material. Once foundation models are deployed,

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ensuring high-quality data is still an issue, as undesirable behavior can still emerge from small subsets of data.

>> Section 5

General-purpose AI Due to their adaptability to a wide range of use-cases, foundation models are sometimes considered to be examples of general-purpose AI. In designing the EU AI Act, the European Parliament has stated that a new wave of general-purpose AI technologies shapes the overall AI ecosystem. The fuller structure of the ecosystem, in addition to the properties of specific general-purpose AI systems, influences the design of AI policy and research. General-purpose AI systems also often appear in people's everyday lives through applications and tools like ChatGPT or DALL-E. Government agencies like EU Parliament have identified regulation of general-purpose AI, such as foundation models, to be a high priority. General-purpose AI systems are often characterized by large size, opacity, and potential for emergence, all of which can create unintended harms. Such systems also heavily influence downstream applications, which further exacerbates the need for regulation. In regards to prominent legislation, a number of stakeholders have pushed for the EU AI Act to include restrictions on general-purpose AI systems, all of which would also apply to foundation models.